"This material contains information affecting the national defense of the United States within the meaning of the espionage laws, title 18, U. S. C., secs 793 and 794, the transmission or revelation of which in any manner to an unauthorized person is prohibited by law."

C-MIL-T-21677(Ord)

21 November 1958
SUPERSEDING
NAVORD OS 5836C
31 March 1955

MILITARY SPECIFICATION

TEST SET MARK 3 MOD 2

DOWNGRADED BY AUTHORITY
OF NAVORDINST 5511.9A
ORD-765 24 MARCH 1972

1. SCOPE

1.1 This specification establishes the requirements for the procurement of the Test Set Mark 3 Mod 2. This test set measures resistance, voltage, and current, and tests the operation of magnetic induction mines.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lockcorner.
PPP-B-636	Boxes, Fiber.

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-B-131	Barrier Material: Water Vaporproof, Flexible.
MIL-V-173	Varnish, Moisture-and Fungus resistant for the Treatment of Communications, Electronic and Associated Electrical Equipment.
MIL-S-6872	Soldering Processes, General Specifications for.

FSC 6625

Source: https://assist.dla.mil -- Downloaded: 20260-19T10:18Z
Check the source to verify that this is the current version before use.

MIL-P-17555(Ships)	Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWINGS

BUREAU OF ORDNANCE

LD 166602	Test Set Mark 3 Mod 2.
LD 284866	Preparation for Delivery for Test Set Cable Assemblies.
1476194	Marking Instructions for Storage and Shipping Containers.

And all documents listed in the above LD's and drawings.

PUBLICATIONS

BUREAU OF ORDNANCE

NAVORD OP 400	General Instructions for Design, Manufacture and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 General requirements.

3.1.1 The Test Set Mark 3 Mod 2 shall be manufactured in accordance with LD 166602 and all documents listed thereon.

3.1.2 The requirements and procedures established in OP 400 shall be complied with as applicable.

Source: https://assist.dla.mil -- Downloaded: 202610-19T10:18Z
Check the source to verify that this is the current version before use.

3.1.3 Preproduction sample. - Unless otherwise specified in the contract or order, a preproduction sample shall be manufactured by the same procedures and processes and at the same location as proposed for full production. This sample will be tested for the purpose of determining compliance with the requirements of this specification, the drawings, and the contract. See 4.1.2.

3.1.4 Interchangeability. - Component parts of assembled units (unless otherwise specified on the applicable drawings), fabricated in strict agreement with their applicable drawings, shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during assembly of the fabricated parts of a unit, selection or fitting is required and changes in dimensions are necessary, no changes in the parts as delineated on applicable drawings shall be made without prior approval of the Bureau of Ordnance.

3.2 Moisture-and-fungus-resistant treatment. - Materials and parts of this test set which are not moisture and fungus resistant shall be treated, before assembly, with a varnish conforming to Military Specification MIL-V-173.

3.3 Requests for deviation. - Requests for deviation from this specification, applicable drawings, specifications, publications, materials, and processes specified herein, shall be as set forth in OP 400.

3.4 Conflicting requirements. - Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or aprointed agent for interpretation and clarification.

3.5 Soldering. - Unless otherwise specified, soldering processes shall conform to the requirements specified in Military Specification MIL-S-6872. Only non-corrosive and non-volatile fluxes shall be used. Rosin shall be used when soldering wires to contacts and terminals. The use of rosin core solder is permitted. Electrical connections shall be mechanically secure and electrically continuous before soldering. Sufficient heat shall be applied during the soldering process to avoid "cold" solder joints.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T10:18Z
Check the source to verify that this is the current version before use.

3.6 Wiring. - All internal wiring shall be cabled wherever possible.

3.7 Performance requirements and product characteristics. - The test set covered by this specification shall meet the following performance requirements and product characteristics.

3.7.1 Insulation resistance. - The insulation resistance between electrical circuits and between each circuit and the chassis shall be not less than 50 megohms when tested as specified in 4.5.3.

3.7.2 Meter, 50 ampere range. - The test set meter shall indicate 10, 20, 30, 40, and 50 amperes ± 1 ampere when connected and tested as specified in 4.5.4.

3.7.3 Voltmeter. - The test set meter shall indicate 5, 10, and 15 volts ± 0.3 volt when connected and tested as specified in 4.5.5.

3.7.4 High ohms circuit. - The test set meter shall indicate 10, 50, 100, 200, and 500 ohms, and be accurate within ± 5 percent, when connected and tested as specified in 4.5.6.

3.7.5 Low ohms circuit. - The test set meter shall indicate 1.0, 2.0, 3.0, 5.0, 10, and 50 ohms, and be accurate within ± 5 percent, when connected and tested as specified in 4.5.7.

3.7.6 Meter, 100 milliampere range. - The test set meter shall indicate 20, 40, 60, 80, and 100 milliamperes on the 0-100 MA scale, and be accurate within ± 2 milliamperes, when connected and tested as specified in 4.5.8.

3.7.7 Meter, 500 milliampere range. - The test set meter shall indicate 100, 200, 300, 400, and 500 milliamperes, (on the 0-500 MA scale), and be accurate within ± 10 milliamperes. Further, with the polarity of external meter connections reversed, the test set meter shall indicate same as external meter within ± 10 milliamperes, for any setting of R1, R2, and R3. With the current off, the meter shall return to 0. Connection and testing shall be as specified in 4.5.9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:18Z
Check the source to verify that this is the current version before use.

3.7.8 Meter, 10 ampere range. - With switches set as specified in 4.5.10, the test set meter shall indicate 10 times that of the external meter reading, or 10 amperes, within ±0.2 amperes. With the current off, the meter will return to 0.

3.8 Workmanship and cleanliness. - The test set shall be free from dust, dirt, metal filings, moisture, solder splashes, chips, and other foreign matter. Necessary precautions shall be taken during manufacture and assembly to insure these conditions.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Government inspection. - The Government inspector will make such inspections as are necessary to determine that all components and assemblies are in accordance with the requirements of the contract, pertinent specifications and drawings. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in OP 400.

4.1.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection will be in accordance with the provisions of MIL-STD-105.

4.1.2 Preproduction sample. - Unless otherwise specified for each contract or order, a representative sample of one test set shall be manufactured using the methods and procedures proposed for full production. The Government inspector shall perform, witness, or participate in such inspections and tests on the sample as are required for full production. Upon conditional acceptance by the inspector, the preproduction sample shall be forwarded, together with a report of inspection, to an activity designated in the contract or by the procuring agency, for the purpose of determining compliance with the requirements of the contract, specifications and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of the tests on preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T10:18Z
Check the source to verify that this is the current version before use.

4.1.2.1 Preproduction sample for a subsequent contract. - The necessity for a preproduction sample will be determined by the procuring agency when production under a new contract by the same contractor at the same location follows the manufacture of satisfactory material as covered by this document.

4.1.3 Lot. - As applied to the Government inspection of units of product, the term "lot" shall mean "inspection lot", i.e. a collection of units of the product used as a basis for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" shall be as determined by the Government inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Contractor's inspection. - Contractor's inspection shall be conducted in accordance with provisions of "Inspection by Contractor" as contained in OP 400.

4.3 Classification of tests. - The inspection and testing of test sets covered by this document shall be classified as follows:

(a) acceptance tests

(b) preproduction tests

4.3.1 Acceptance tests. - Acceptance tests for acceptance of the test set covered by this specification shall consist of tests to determine compliance with all requirements specified in this document.

4.3.2 Preproduction tests. - Preproduction tests are those performed on samples submitted in accordance with 4.1.2 to determine that the lot and production meet all the requirements of this specification. Acceptance shall be based on no defects in a sample.

4.4 Test equipment.

4.4.1 Test and inspection equipment and facilities. - Unless otherwise specified, the manufacturer shall furnish and maintain all necessary test equipment, facilities and personnel for performing all acceptance tests under the

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:18Z
Check the source to verify that this is the current version before use.

supervision of the Government inspector. The test equipment shall be adequate in quantity, and unless definite requirements are specified, they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.4.1.1 Equipment required for performing these tests is as follows:

(a) Variable d-c current source, 0-50 amperes as obtained from a 6 volt, 100 ampere-hour or better storage battery, with a one ohm, 50 watt, rheostat connected across the battery.

(b) d-c ammeter, 0-10 amperes, accuracy within ±0.5 percent.

(c) d-c ammeter, 0-50 amperes, accuracy within ±0.5 percent.

(d) Variable d-c voltage source, 0-15 volts capable of delivering 50 MA with a ripple voltage not greater than 2 percent.

(e) d-c voltmeter, 0-15 volts, accuracy within ±0.5 percent.

(f) d-c milliammeter, 0-100 milliamperes, accuracy within ±0.5 percent.

(g) d-c milliammeter, 0-500 milliamperes, accuracy within ±0.5 percent.

(h) Variable resistance, 0-500 ohms, 1 watt, accuracy within ±1 percent.

(i) Megohmmeter, 500 volts d-c. Range 10.0-100 megohms, accuracy within ±10 percent.

4.5 Tests.

4.5.1 Test conditions. - Unless otherwise specified, tests shall be performed at 80° ± 10° F and ambient relative humidity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:18Z
Check the source to verify that this is the current version before use.

4.5.2 Visual inspection. - The test set and its component parts shall be inspected to verify that the materials, design, construction, physical dimensions, soldering, and workmanship are in accordance with this document and the drawings.

4.5.3 Insulation resistance. - The test set shall be subjected to a conditioning period of 24 hours at a temperature of 77° ±2° F, and a relative humidity of 75 ±5 percent. A test potential of 500 ± 20 volts d-c shall then be applied as specified in 3.7.1. The insulation resistance shall be not less than 50 megohms.

4.5.4 Meter, 50 ampere range test.

(a) Connect 0 to 50 ampere variable d-c current source (item 4.4.1.1 (a)) and 0 to 50 external d-c ammeter (item 4.4.1.1 (c)) in series between test set terminals 50 AMPS., observing polarity.

(b) Set SELECTOR SW. at 1, and OPER-ADJ switch at ADJ. Adjust the variable current source so that the test set meter indicates 10, 20, 30, 40, and 50 amperes. When compared with the readings of the external meter, the test set meter shall be accurate within ± 1 ampere.

4.5.5 Voltmeter test.

(a) Connect cable CA-716 to receptacle on test set.

(b) Connect 0 to 15 volts variable d-c voltage source (item 4.4.1.1 (d)) together with external d-c voltmeter (item 4.4.1.1 (e)) across leads 1 and 2 of CA-716, lead 1 being positive. Adjust the variable voltage source so that the test set meter indicates 5, 10, and 15 volts. When compared with the readings of the external meter the test set meter shall be accurate within ± 0.3 volt.

4.5.6 High ohms circuit test.

(a) Set SELECTOR SW at 4, OPER-ADJ switch at ADJ, and place jumper across OHMS terminals.

(b) Turn OHMMETER ZERO ADJ until test set meter indicates zero on 0-2000 ohms scale. Remove jumper.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:18Z
Check the source to verify that this is the current version before use.

(c) Connect 0 to 500 ohms variable resistance (item 4.4.1.1 (h)) across OHMS terminals, and adjust it so that test set meter indicates 10, 50, 100, 200, and 500 ohms. When compared with the actual resistance applied, the test set meter shall be accurate within ± 5 percent.

4.5.7 Low ohms circuit test.

(a) Set SELECTOR SW at 5, and OPER-ADJ switch at ADJ.

(b) Turn OHMMETER ZERO ADJ until test set meter indicates infinity (oo) on the 0-50 OHMS scale.

(c) Connect 0 to 500 ohms variable resistance (item 4.4.1.1 (h)) across OHMS terminals, and adjust it so that test set meter indicates 1.0, 2.0, 3.0, 5.0, 10, and 50 ohms. When compared with the actual resistance applied, the test set meter shall be accurate within ± 5 percent.

NOTE: When this test is completed, set SELECTOR SW at any position other than 5, to eliminate excessive drain of test set battery.

4.5.8 Meter, 100 milliampere range test.

(a) Connect the 0 to 100 external milliammeter (item 4.4.1.1 (f)) across the EXCITER COIL terminals.

(b) Set SELECTOR SW at 2 and OPER-ADJ switch at ADJ.

(c) Hold REV SW at position which causes external meter to indicate in positive direction. Adjust R1, R2, and R3 on test set panel so that meter indicates 20, 40, 60, 80, and 100 milliamperes on 0-100 MA scale. When compared with the readings of the external meter, the test set meter shall be accurate within ± 2 milliamperes.

4.5.9 Meter, 500 milliampere range test.

(a) Connect the 0 to 500 external milliammeter (item 4.4.1.1 (g)) across the EXCITER COIL terminals.

(b) Set SELECTOR SW at position 3 and OPER-ADJ switch at ADJ.

Source: https://assist.dla.mil -- Downloaded: 202610-19T10:18Z
Check the source to verify that this is the currentversion before use.

(c) Hold REV SW at position which causes external meter to indicate in positive direction. Adjust R1 and R2 so that test set meter indicates 100, 200, 300, 400, and 500 milliamperes on the 0-500 MA scale. When compared with readings of the external meter, the test set meter shall be accurate within ± 10 milliamperes.

(d) Release REV SW. Test set meter and external meter shall indicate zero.

(e) Reverse polarity of external meter connections, and hold REV SW at position opposite to that in step (c). Test set meter shall indicate same as external meter within ± 10 milliamperes, for any setting of R1, R2, and R3.

(f) Set OPER-ADJ switch at OPER. Test set meter shall indicate zero.

(g) Release REV SW. Test set meter and external meter shall indicate zero.

4.5.10 Meter, 10 ampere range test.

(a) Connect CA-716 cable to receptacle on test set.

(b) Connect 0 to 50 ampere variable d-c current source (item 4.4.1.1 (a)) and 0-10 ampere external d-c ammeter (item 4.4.1.1 (b)) in series between leads 3 and 5 of CA-716, lead 5 being negative.

(c) Set 10 AMP-1 AMP switch at 10 AMP, and OPER-ADJ switch at OPER.

(d) Set SW B at up (on) position. Adjust the variable current source so that the test set meter indicates 2, 4, 6, 8, and 10 amperes on the 0-10 AMPS scale. When compared with the external meter, the test set meter shall be accurate within ± 0.2 ampere.

(e) Adjust variable current source so that external meter indicates 1 ampere.

(f) Set 10 AMP-1 AMP switch at 1 AMP. The test set meter shall indicate 10 times external meter, or 10 amperes, within 0.2 ampere.

Source: https://assist.dla.mil -- Downloaded: 202610-19T10:18Z
Check the source to verify that this is the current version before use.

(g) Set SW B at down position (off). Test set meter and external meter shall indicate zero.

(h) Repeat step (b), except use lead 4 of CA-716 instead of lead 5, as the negative lead.

(i) Set 10 AMP-1 AMP switch at 10 AMP.

(j) Set SW A and SW B at up position and adjust variable current source so that the external meter indicates 10 amperes. Test set meter shall also indicate 10 amperes within ± 0.2 ampere.

(k) Set OPER-ADJ switch at ADJ. Test set meter shall indicate zero.

(l) Set 10 AMP-1 AMP switch at center position. External meter shall indicate zero.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. - In addition to the following, Level A packaging shall be in accordance with Specification MIL-P-17555.

5.1.1.1 Cleaning, drying, and preservative application. (Not applicable).

5.1.1.2 Unit packaging. - Prior to unit packaging, all test set cables shall be packaged in accordance with Bureau of Ordnance LD No. 284866. Each test set shall be packaged in a Type I, Class 1, Style RSC corrugated fiberboard box, having a minimum Mullen test of 275 pounds, conforming to the requirements of Federal Specification PPP-B-636. Sufficient fillers and folded corner pads, fabricated from the same box material, shall be utilized within the carton to provide a tight pack and to prevent direct contact between protrusions of the test set case and the walls of the fiberboard box.

5.1.1.2.1 The unit package, prepared as described above, shall be further packaged Method IA-14 conforming to the requirements of Military Specification MIL-P-116. The inner carton with all corners blunted shall be inserted into a water-vaporproof bag fabricated from Class I barrier material

Source: https://assist.dla.mil -- Downloaded: 20260-19T10:18Z
Check the source to verify that this is the current version before use.

conforming to the requirements of Military Specification MIL-B-131, and closure heat-sealed. The outer carton shall be a Class 2, Style RSC fiberboard box conforming to the requirements of Federal Specification PPP-B-636, V5c or V5s compliance symbol.

5.1.1.3 Intermediate package. - (Not applicable)

5.1.2 Level B. - (Not applicable)

5.1.3 Level C.

5.1.3.1 Cleaning, drying, and preservative application. (Not applicable.)

5.1.3.2 Unit packaging. - Unit packaging shall be in accordance with Method III of Military Specification MIL-P-116. The unit package shall contain one (1) test set.

5.1.3.3 Intermediate package. - (Not applicable)

5.2 Packing.

5.2.1 Level A. - In addition to the following, Level A packaging shall be in accordance with Military Specification MIL-P-17555.

5.2.1.1 Exterior containers. - Each test set, individually packaged as specified above for Level A protection, shall be packed in an overseas type wood-cleated plywood box or in a Class 2, Style 2, 2 1/2, 3 or 4 nailed wood box conforming to the requirements of Federal Specifications PPP-B-601 or PPP-B-621, respectively.

5.2.1.1.1 When a nailed wood box is selected to be used for shipment, box lumber shall conform to Group I, II or III woods of Federal Specification PPP-B-621. The lumber shall be of one (1) inch nominal thickness and surfaced on both sides.

5.2.2 Level B. - In addition to the following, Level B packaging shall be in accordance with Military Specification MIL-P-17555.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:18Z
Check the source to verify that this is the current version before use.

5.2.2.1 Exterior containers. - Each test set, individually packaged as specified above for Level A protection, shall be packed in a domestic type wood-cleated plywood box or in a Class 1, Style 2, 2 1/2, 3 or 4 nailed wood box conforming to the requirements of Federal Specifications PPP-B-601 or PPP-B-621, respectively.

5.2.3 Level C.

5.2.3.1 Exterior containers. - Each test set, packaged in accordance with 5.1.3.2, shall be packed in a manner which will provide protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Special marking. - The following precautionary marking shall be stenciled or stamped on the top, one (1) side and one (1) end of the shipping container:

DELICATE INSTRUMENT
HANDLE WITH CARE

5.3.2 Normal marking. - In addition to any special marking required by the contract or order, all marking shall be in accordance with Military Standard MIL-STD-129, including serial numbers.

5.3.2.1 For detailed marking requirements, see Bureau of Ordnance Drawing number 1476194.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the title, number, and date of this specification, and exceptions to this specification, applicable drawings and other documents. Procurement documents shall also specify the desired level of packaging and packing. The attention of the contracting officer is invited to the Quality Assurance Provisions and options in Specification MIL-P-116, and to paragraph 6.1.1 of this document.

Source: https://assist.dla.mil -- Downloaded: 20260-19T10:18Z
Check the source to verify that this is the current version before use.

6.1.1 Criteria for use of proper level of preservation, packaging and packing shall be as follows:

For Level A. - This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level B. - This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for 6 months or less or in domestic locations for an indefinite period.

For Level C. - This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever: and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T10:18Z
Check the source to verify that this is the current version before use.